**Faculty of Engineering and Technology**
**Electrical and Computer Engineering Department**

# Operating System Concepts
# ENCS3390

## Programming Task 2


Prepared by:

| | | |
|---|---|---|
| **Joud Thaher** | **1221381** | **sec2** |
| **Labiba Sharia** | **1220228** | **sec1** |


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## Test cases without deadlock

We began by testing simple scenarios and gradually moved on to more complex ones, including the test cases outlined in the project document. The scheduling method used is non-pre-emptive priority scheduling algorithm with round robin

1. Test Case provided in the project document

```
0 0 1 CPU{R[1],50,F[1]}
1 5 1 CPU{20} IO{30} CPU{20,R[2],30,F[2],10}
```

```
Gantt Chart:
=================================================================
Time:     0         50        70       100       160
=================================================================
Proc:     P0        P1        -         P1        -
=================================================================

Process Metrics:
==========================================================
   PID    |  Arrival   | Completion | Turnaround |  Waiting
----------------------------------------------------------
    0     |     0      |     50     |     50     |     0
    1     |     5      |    160     |    155     |    45
----------------------------------------------------------

Average Turnaround Time: 102.50
Average Waiting Time: 22.50
```

2. Test Case provided in the test cases document.

```
0 0 0 CPU{10} IO{30} CPU{10}
1 0 0 CPU{20}
```

```
Gantt Chart:
=================================================================================
Time:     0         10        30        40        50
=================================================================================
Proc:     P0        P1        -         P0        -
=================================================================================

Process Metrics:
==========================================================
   PID    |  Arrival   | Completion | Turnaround |  Waiting
----------------------------------------------------------
    0     |     0      |     50     |     50     |     0
    1     |     0      |     30     |     30     |    10
----------------------------------------------------------

Average Turnaround Time: 40.00
Average Waiting Time: 5.00
```

3. First Designed Test Case: There are 5 processes in total. Processes 1 and 2 share the same arrival time and priority, so they will be scheduled using Priority Round-Robin (RR). Similarly, processes 4 and 5 also share the same arrival time and priority, so they will also use Priority Round-Robin (RR).

```
1 0 2 CPU{8} IO{4} CPU{6}
2 0 2 CPU{10} IO{3} CPU{5}
3 1 3 CPU{12} IO{6} CPU{8}
4 2 1 CPU{7} IO{5} CPU{9}
5 2 1 CPU{9} IO{4} CPU{7}
```

With quantum = 4

```
Gantt Chart:
==================================================================================
Time:    0       4       8       12      15      20      29      36      40      44      50      56      61      73      79      87
==================================================================================
Proc:    P1      P4      P5      P4      P5      P4      P5      P2      P1      P2      P1      P2      P3      -       P3      -
==================================================================================

Process Metrics:
=========================================================
  PID   | Arrival  | Completion | Turnaround |  Waiting
---------------------------------------------------------
   1    |    0     |    56      |    56      |    38
   2    |    0     |    61      |    61      |    43
   3    |    1     |    87      |    86      |    60
   4    |    2     |    29      |    27      |    6
   5    |    2     |    36      |    34      |    14
---------------------------------------------------------

Average Turnaround Time: 52.80
Average Waiting Time: 32.20
```

With quantum = 10

```
Gantt Chart:
==================================================================================================
Time:    0       8       15      24      33      40      50      56      61      73      79      87
==================================================================================================
Proc:    P1      P4      P5      P4      P5      P2      P1      P2      P3      -       P3      -
==================================================================================================

Process Metrics:
=========================================================
  PID   | Arrival  | Completion | Turnaround |  Waiting
---------------------------------------------------------
   1    |    0     |    56      |    56      |    38
   2    |    0     |    61      |    61      |    43
   3    |    1     |    87      |    86      |    60
   4    |    2     |    33      |    31      |    10
   5    |    2     |    40      |    38      |    18
---------------------------------------------------------

Average Turnaround Time: 54.40
Average Waiting Time: 33.80
```

4. Second Designed Test Case: There are 4 processes in total, 3 of them have the same arrival time and priority, so they will be scheduled using Priority Round-Robin (RR).

   With quantum = 5

```
0 0 1 CPU{9} IO{5} CPU{11}
1 0 1 CPU{6}
2 0 1 CPU{10} IO{2} CPU{5}
3 7 0 CPU{13}
```

```
Gantt Chart:
==================================================================================================
Time:     0        5        10       23       28       32       33       38       49       54
==================================================================================================
Proc:     P0       P1       P3       P2       P0       P1       P2       P0       P2       -
==================================================================================================

Process Metrics:
==========================================================
   PID    |  Arrival   | Completion | Turnaround |  Waiting
----------------------------------------------------------
    0     |     0      |     49     |     49     |     24
    1     |     0      |     33     |     33     |     27
    2     |     0      |     54     |     54     |     37
    3     |     7      |     23     |     16     |     3
----------------------------------------------------------

Average Turnaround Time: 38.00
Average Waiting Time: 22.75
```

5. Third Designed Test Case

   With quantum = 5

```
0 0 0 CPU{10} IO{15} CPU{20} IO{10} CPU{15}
1 0 0 CPU{15} IO{10} CPU{10}
```

```
Gantt Chart:
==================================================================================================
Time:     0        5        10       15       25       30       50       60       75
==================================================================================================
Proc:     P0       P1       P0       P1       -        P0       P1       P0       -
==================================================================================================

Process Metrics:
==========================================================
   PID    |  Arrival   | Completion | Turnaround |  Waiting
----------------------------------------------------------
    0     |     0      |     75     |     75     |     5
    1     |     0      |     60     |     60     |     25
----------------------------------------------------------

Average Turnaround Time: 67.50
Average Waiting Time: 15.00
```

## Test cases with deadlock

The detection method employs the wait-for cycle approach, and recovery is achieved through process termination.

1. Test Case provided in the test cases document: q=10

```
0 0 1 CPU{R[1], 15, R[2], 10, F[1], F[2]}
1 0 1 CPU{5, R[2], R[1], 10, F[2], F[1]}
2 0 0 CPU{50}
```

```
Gantt Chart:
==========================================================================
Time:     0         50        60        70        80        85        90
==========================================================================
Proc:     P2        P0        P1        P0        P1        P0        -
==========================================================================

Process Metrics:
==========================================================
   PID    |  Arrival   | Completion | Turnaround |  Waiting
----------------------------------------------------------
    0     |     0      |     90     |     90     |     65
    1     |     0      |     85     |     85     |     70
    2     |     0      |     50     |     50     |     0
----------------------------------------------------------

Average Turnaround Time: 75.00
Average Waiting Time: 45.00
```

2. First Designed Test Case

```
0 0 1 CPU{R[1], 10, R[2], 5, F[1], F[2]}
1 0 1 CPU{R[2], 10, R[3], 5, F[2], F[3]}
2 0 1 CPU{R[3], 10, R[1], 5, F[3], F[1]}
```

```
Gantt Chart:
================================================================================
Time:     0         10        20        30        35        40        45
================================================================================
Proc:     P0        P1        P2        P0        P1        P2        -
================================================================================

Process Metrics:
==========================================================
   PID    |  Arrival   | Completion | Turnaround |  Waiting
----------------------------------------------------------
    0     |     0      |     35     |     35     |     20
    1     |     0      |     40     |     40     |     25
    2     |     0      |     45     |     45     |     30
----------------------------------------------------------

Average Turnaround Time: 40.00
Average Waiting Time: 25.00
```

For the test cases with deadlock, unfortunately the code didn't work for both the scheduling and deadlock correctly so we made a separate code for deadlock covering the test cases we just provided:

Note that the pictures will cover the output of different test cases shown in the code.

**Test Case 1:**

```
=== Test Case 1: Basic Deadlock Scenario ===

Step 1: P0 acquires R1

Resource Allocation Graph:
Allocations (Resource → Process):
R1 → P0

Waiting (Process → Resource):

Step 2: P1 acquires R2

Resource Allocation Graph:
Allocations (Resource → Process):
R1 → P0
R2 → P1

Waiting (Process → Resource):
```

```
Step 3: P0 requests R2

Resource Allocation Graph:
Allocations (Resource → Process):
R1 → P0
R2 → P1

Waiting (Process → Resource):
P0 → R2

Step 4: P1 requests R1

Deadlock Recovery: Terminated P1 to resolve deadlock

Resource Allocation Graph:
Allocations (Resource → Process):
R1 → P0
R2 → P0

Waiting (Process → Resource):

Wait-For Graph:
```

**Test Case 2:**

```
=== Test Case 2: Time Quantum Scenario ===

Step 1: P0 acquires R1

Step 2: P0 requests R2

Step 3: P1 acquires R2

Step 4: P1 requests R1

Resource Allocation Graph:
Allocations (Resource → Process):
R1 → P0
R2 → P0

Waiting (Process → Resource):
P1 → R1
P1 → R2

Wait-For Graph:
P1 → P0
```

**Test Case 3:**

```
=== Test Case 3: Circular Wait Scenario ===

Step 1: P0 acquires R1

Step 2: P1 acquires R2

Step 3: P2 acquires R3

Step 4: P0 requests R2

Step 5: P1 requests R3

Step 6: P2 requests R1

Deadlock Recovery: Terminated P1 to resolve deadlock

Resource Allocation Graph:
Allocations (Resource → Process):
R1 → P0
R2 → P0
R3 → P2

Waiting (Process → Resource):
P2 → R1
```

```
Deadlock Recovery: Terminated P1 to resolve deadlock

Resource Allocation Graph:
Allocations (Resource → Process):
R1 → P0
R2 → P0
R3 → P2

Waiting (Process → Resource):
P2 → R1

Wait-For Graph:
P2 → P0
```

**Test Case 4:**

```
=== Test Case 4: Priority-based Scenario ===

Step 1: P0 acquires R2

Step 2: P1 acquires R1

Step 3: P2 requests R1 and R2

Resource Allocation Graph:
Allocations (Resource → Process):
R2 → P0
R1 → P1

Waiting (Process → Resource):
P2 → R1
P2 → R2

Wait-For Graph:
P2 → P0
P2 → P1
```